Unit 3: Knowledge Representation and Probabilistic Reasoning

SubUnit 3.5: Probabilistic Reasoning and Bayes' Theorem

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The Need for Probabilistic Reasoning

Problem with Pure Logic:

- Real world is **uncertain** and **incomplete**
- Cannot always know everything with certainty
- Logical reasoning: binary (true/false) - too rigid

Real-world Scenarios

- Medical diagnosis: Symptoms suggest possible diseases (not certain)
- Weather prediction: Probability of rain based on atmospheric conditions
- Spam filtering: Email characteristics indicate likelihood of spam
- Robot navigation: Sensor readings have noise and uncertainty

Solution: Probability Theory

Probability theory provides a **rigorous framework** for reasoning under uncertainty, allowing agents to make **rational decisions** with incomplete information.

Probability: Basic Concepts

Probability measures the likelihood of an event occurring

Key Definitions:

- **Sample Space** (Ω): Set of all possible outcomes
- **Event**: Subset of sample space
- **Probability** P(A): Number between 0 and 1
  - P(A) = 0: Event A never occurs
  - P(A) = 1: Event A always occurs
  - 0 ¡ P(A) ¡ 1: Event A occurs with some uncertainty

Fundamental Rules:

1. $0 \leq P(A) \leq 1$ for any event A
2. $P(\Omega) = 1$ (something must happen)
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Example

Die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$, P(rolling 3) $= \frac{1}{6} \approx 0.167$

Conditional Probability

Definition

Conditional Probability $P(A|B)$ is the probability of event A occurring given that event B has occurred.

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) > 0$$

Medical Example

- P(Disease) $= 0.01$ (1% of population has disease)
- $P(PositiveTest|Disease) = 0.95$ (test is 95% accurate if disease present)
- $P(PositiveTest|NoDisease) = 0.05$ (5% false positive rate)

Question: If test is positive, what's probability of having disease?

Key Insight:

- Conditional probability updates our beliefs based on new evidence
- Essential for reasoning with partial information

Reference: Russell & Norvig (2020), Section 13.2

Joint and Marginal Probability

Joint Probability: P(A, B) or P(A ∩ B)

- Probability that both A and B occur
- **Product Rule:** $P(A, B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$

Marginal Probability: Obtained by summing over other variables

- **Marginalization:** $P(A) = \sum_b P(A, B = b)$

Example: Weather and Traffic

Rain	Traffic	Joint P(R,T)	
Yes	Heavy	0.3	
Yes	Light	0.1	P(Rain=Yes) = 0.4
No	Heavy	0.2	
No	Light	0.4	P(Rain=No) = 0.6
	P(Heavy)=0.5	P(Light)=0.5	Total = 1.0

Reference: Russell & Norvig (2020), Section 13.2

Bayes' Theorem: The Foundation

Bayes' Theorem

$$\boxed{P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}}$$

Components:

- $P(H|E)$: Posterior probability - What we want to know
- $P(E|H)$: Likelihood - How likely is evidence given hypothesis
- **P(H)**: Prior probability - Initial belief about hypothesis
- **P(E)**: Evidence probability - Normalizing constant

Expanded Form (using total probability):

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E|H) \cdot P(H) + P(E|\neg H) \cdot P(\neg H)}$$

Key Idea: Update beliefs based on new evidence!

Reference: Russell & Norvig (2020), Section 13.3

Bayes' Theorem: Medical Diagnosis Example

Problem

A disease affects 1% of population. A test is 95% accurate for those with disease (sensitivity) and has 5% false positive rate.

If you test positive, what's the probability you have the disease?

Given Information:

- P(Disease) $= 0.01$, P(No Disease) $= 0.99$
- $P(Positive|Disease) = 0.95$
- $P(Positive|NoDisease) = 0.05$

Medical Diagnosis Example

Apply Bayes' Theorem:

$$
\begin{aligned}
P(\text{Disease}|\text{Positive}) &= \frac{P(\text{Positive}|\text{Disease}) \cdot P(\text{Disease})}{P(\text{Positive})} \\
&= \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} \\
&= \frac{0.0095}{0.0095 + 0.0495} = \frac{0.0095}{0.059} \\
&\approx \mathbf{0.161 \text{ or } 16.1\%}
\end{aligned}
$$

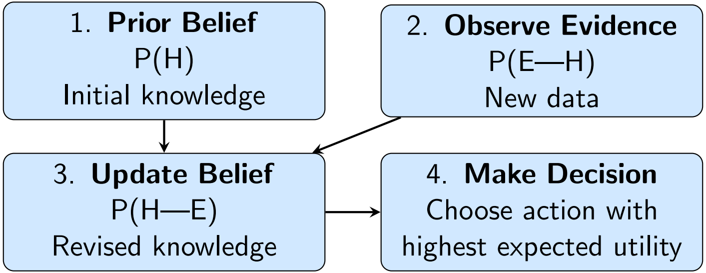
Surprising Result!

Even with a positive test, probability of disease is only 16%! This is due to low prior probability (base rate).

Reference: Russell & Norvig (2020), Example 13.5

Bayesian Inference: The Complete Process

General Framework for Reasoning Under Uncertainty:



Iterative Process:

- Today's posterior becomes tomorrow's prior
- Continuously update beliefs as new evidence arrives
- Converges to truth with sufficient evidence

Reference: Russell & Norvig (2020), Section 13.3

Often we have multiple competing hypotheses

General Form:

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{\sum_j P(E|H_j) \cdot P(H_j)}$$

Robot Localization Example

Robot could be in one of 3 rooms: A, B, C

- **Prior:** P(A)=0.5, P(B)=0.3, P(C)=0.2
- **Evidence:** Robot sees a red door
- **Likelihood:** $P(RedDoor|A)$=0.8, $P(RedDoor|B)$=0.1, $P(RedDoor|C)$=0.3

Bayesian Inference with Multiple Hypotheses Cont...

Robot Localization Example

Calculate Posteriors:

$$P(A|\text{Red}) = \frac{0.8 \times 0.5}{0.8 \times 0.5 + 0.1 \times 0.3 + 0.3 \times 0.2} = \frac{0.4}{0.49} \approx 0.816$$

$$P(B|\text{Red}) = \frac{0.03}{0.49} \approx 0.061 \quad P(C|\text{Red}) = \frac{0.06}{0.49} \approx 0.122$$

Conclusion: Robot is most likely in room A (81.6% probability)

Reference: Russell & Norvig (2020), Section 13.4

Application: Naive Bayes Classifier

Problem: Classify objects based on multiple features

Naive Bayes Assumption: Features are **conditionally independent** given the class

$$P(C|F_1, F_2, \ldots, F_n) \propto P(C) \prod_{i=1}^{n} P(F_i|C)$$

Email Spam Classification

Task: Is email spam or not spam?

Features: Words in email (F="free", F="winner", F="meeting")

$$\begin{aligned} P(\text{Spam}|F_1, F_2, F_3) &\propto P(\text{Spam}) \cdot P(F_1|\text{Spam}) \cdot P(F_2|\text{Spam}) \cdot P(F_3|\text{Spam}) \\ &= 0.3 \times 0.8 \times 0.7 \times 0.1 = 0.0168 \end{aligned}$$

$$P(\text{Not Spam}|F_1, F_2, F_3) \propto 0.7 \times 0.1 \times 0.05 \times 0.4 = 0.0014$$

Decision: $0.0168 > 0.0014 \rightarrow$ Classify as Spam

Bayesian Networks: Graphical Models

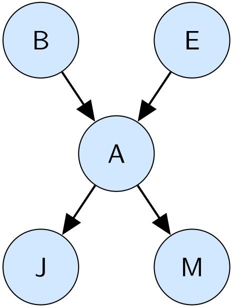
Challenge: Representing complex dependencies efficiently

Bayesian Network:

- Directed Acyclic Graph (DAG)
- Nodes = Random variables
- Edges = Dependencies
- Compact representation

Advantages:

- Efficient storage
- Modular structure
- Exploits conditional independence
- Enables efficient inference



Alarm Example

B: Burglary, E: Earthquake, A: Alarm, J: John Calls, M: Mary Calls

Reference: Russell & Norvig (2020), Chapter 14

Probabilistic Reasoning: Strengths and Challenges

Advantages:

- **Principled framework**: Solid mathematical foundation
- **Handles uncertainty**: Natural representation of incomplete knowledge
- **Optimal decisions**: Maximizes expected utility
- **Learning from data**: Can estimate probabilities
- **Combines evidence**: Integrates multiple information sources
- **Explains reasoning**: Transparent inference process

Challenges:

- **Knowledge acquisition**: Need prior probabilities
- **Computational cost**: Inference can be expensive
- **Independence assumptions**: May not hold in practice
- **Large domains**: Exponential number of parameters
- **Continuous variables**: Discretization issues

Solutions:

- Approximate inference
- Structure learning
- Sampling methods

Summary: Key Concepts and Formulas

Essential Formulas:

1. **Conditional Probability:**
$$P(A|B) = \frac{P(A, B)}{P(B)}$$
2. **Product Rule:**
$$P(A, B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$
3. **Bayes' Theorem:**
$$\boxed{P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}}$$
4. **Total Probability:**
$$P(E) = \sum_i P(E|H_i) \cdot P(H_i)$$

Key Takeaways

- Probability theory enables reasoning under uncertainty
- Bayes' theorem updates beliefs based on evidence (Prior $\rightarrow$ Posterior)
- Conditional independence simplifies computation (Naive Bayes) **Applications:** Document classification, medical diagnosis, recommendation systems
- Foundation for modern AI: diagnosis, filtering, prediction, learning

Bottom Line

Probabilistic reasoning is essential for AI systems operating in uncertain, real-world environments

References: Russell & Norvig (2020), Ch 13; Rich et al. (2009), Ch 8